

PWM BASED AUTOMATIC BRAKING SYSTEM

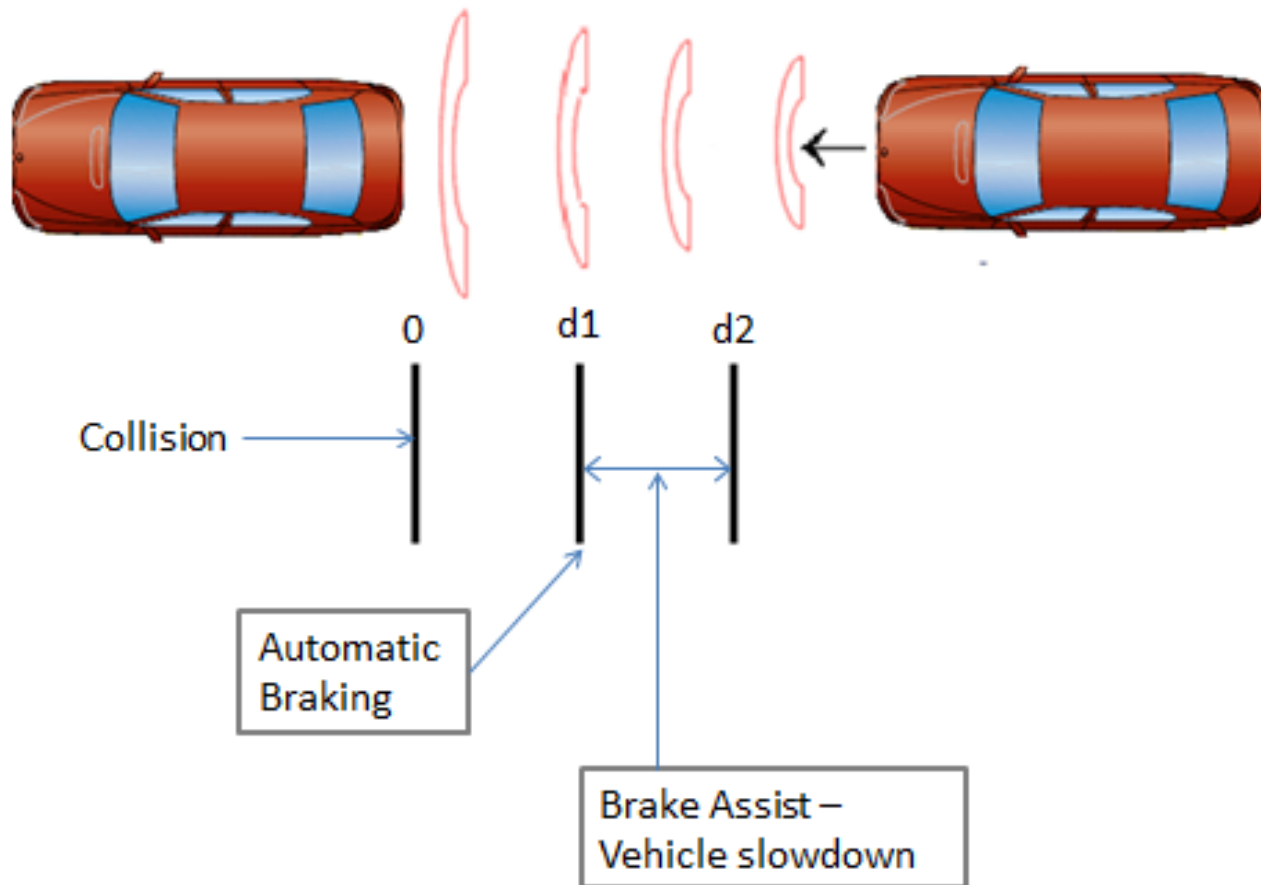
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INTRODUCTION

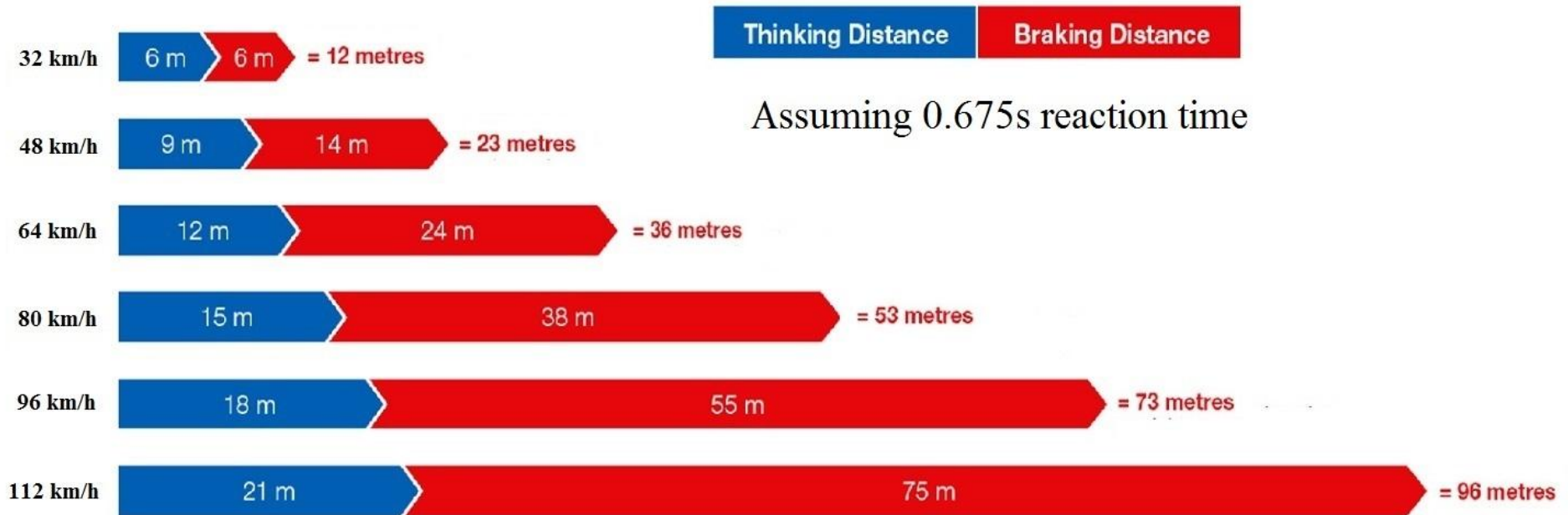
- To sense an imminent collision with another vehicle, person or obstacle.
- To slow the vehicle or apply brakes without any driver input
- Emergency braking can be automatically applied to avoid the collision or at least to reduce its effects.

- United Nations Economic Commission for Europe(UNECE) has announced that this kind of system will become mandatory for new heavy vehicles .
- A recent study suggests that if all cars feature the system, it will reduce accidents by up to 27 percent and save up to 8000 lives per year.

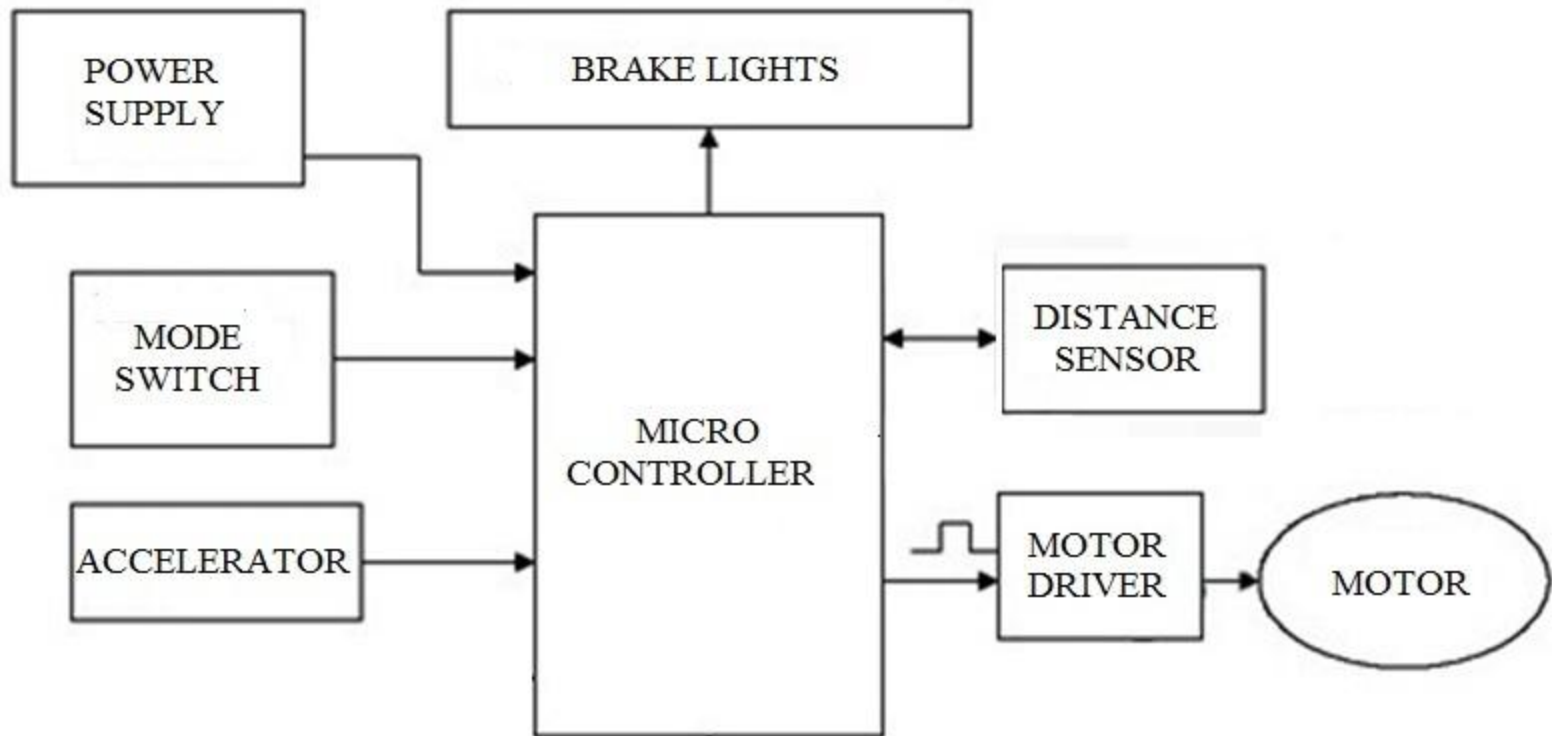


STOPPING DISTANCE [5]

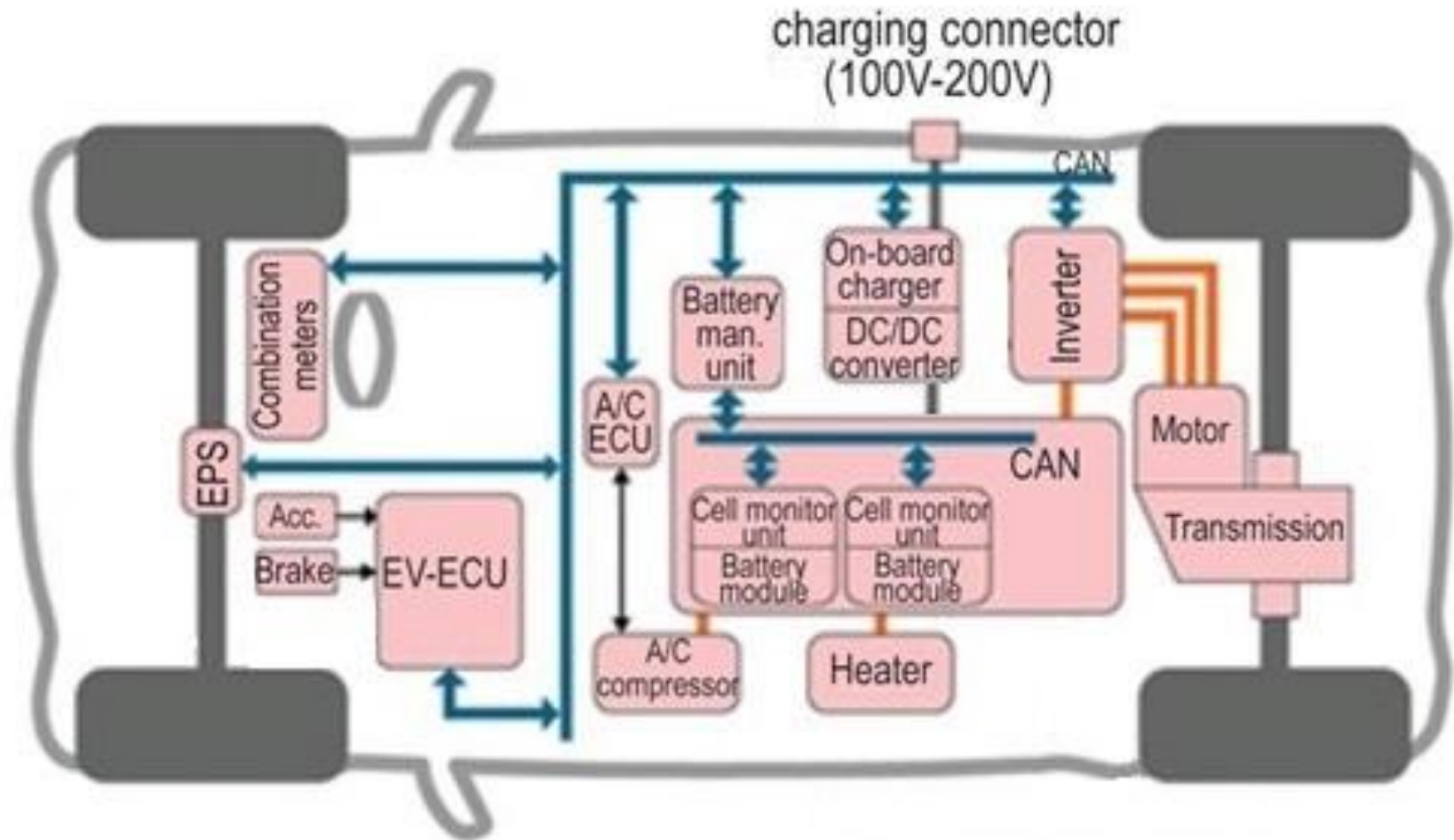
TYPICAL STOPPING DISTANCE



BLOCK DIAGRAM



EV SYSTEM CONFIGURATION [6]



DISTANCE SENSORS

LASER SENSORS



Allen Bradley



SICK

LiDAR

(Light Detection And Ranging)



LiDAR module by SICK

RADAR

(Radio Detection And Ranging)



R-Gage



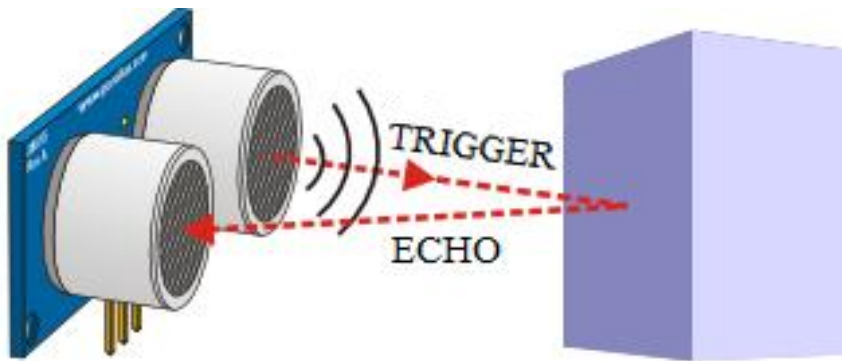
Bosch

STEREO CAMERA



ULTRASONIC SENSOR (HC-SR04)

- Ultrasonic ranging module HC - SR04 provides 4cm - 400cm.
- Ranging accuracy can reach to 3mm.
- The module includes ultrasonic transmitter, receiver and control circuit.

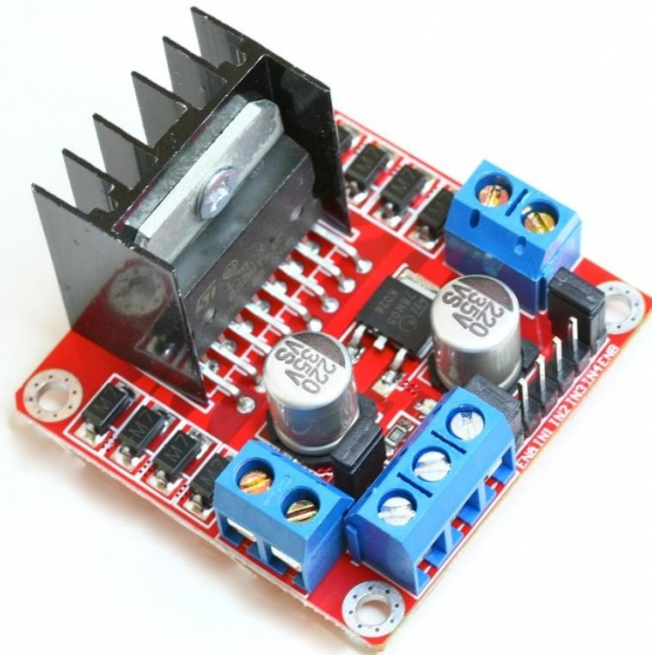


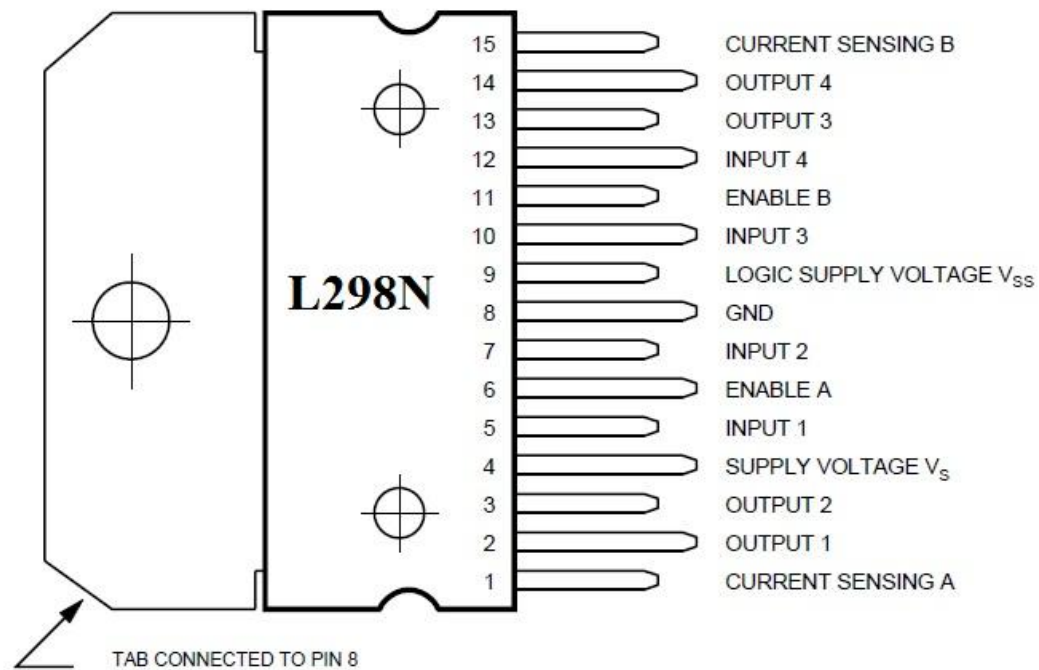
COMPARISON

TYPE	ADVANTAGE	DISADVANTAGE	COST
LASER	SIMPLE	INACCURATE	MODERATE
LIDAR	ACCURATE	COMPLEX WORKING	HIGH
RADAR	HIGH RANGE	HIGH POWER	HIGH
STEREO CAMERA	OBSTACLE DIMENSIONS CAN BE FOUND	AFFECTED BY VISIBILITY CONDITIONS	MODERATE
ULTRASONIC	NOT AFFECTED BY VISIBILITY CONDITIONS	INACCURATE	LOW

L298N MOTOR DRIVER

- Dual full-bridge driver to drive inductive loads such as relays, solenoids, DC and stepping motors
- Integrated monolithic circuit in a 15- pin package.
- Operating Supply Voltage up to 46 V
- Total DC Current Up To 4 A
- Overtemperature Protection

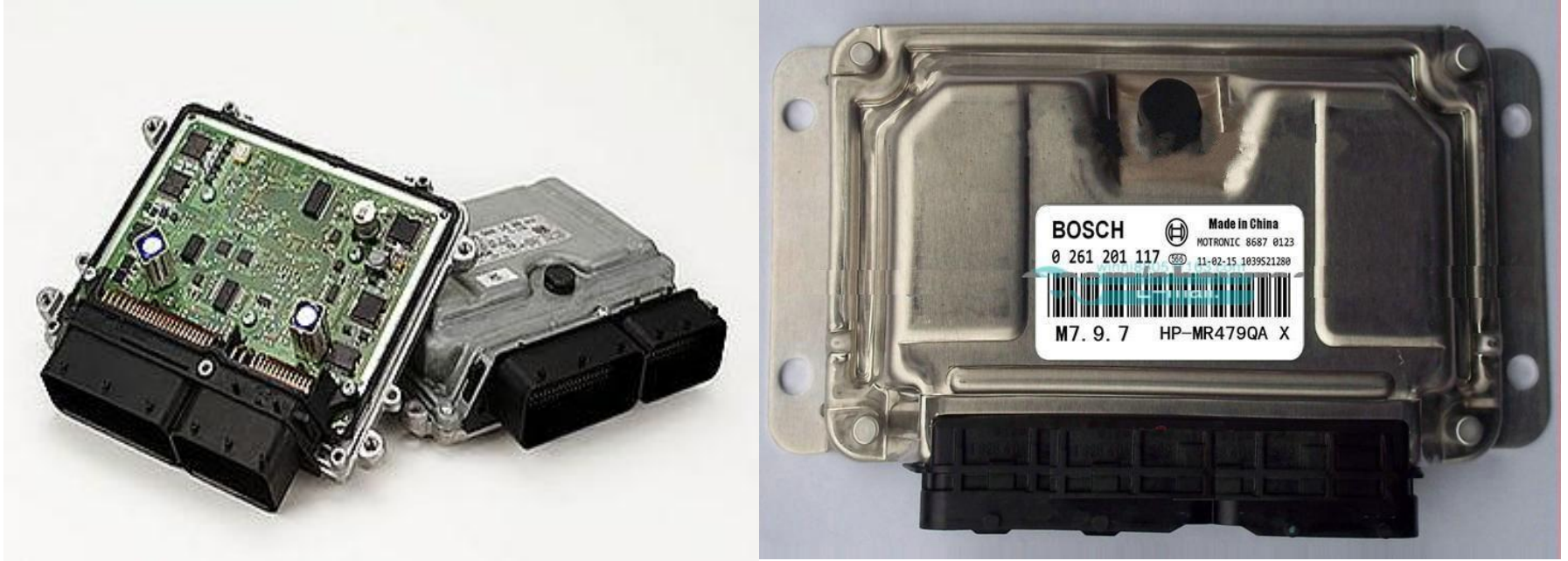




L298N OPERATING MODES

IN1	IN2	EN	FUNCTION
1	0	1	FORWARD
0	1	1	REVERSE
1	1	1	BRAKING
0	0	1	BRAKING
X	X	0	FREE RUNNING STOP

ECU

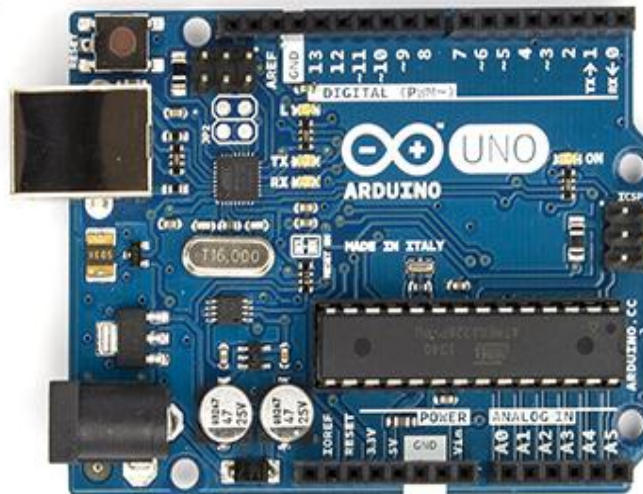


BOSCH ECU

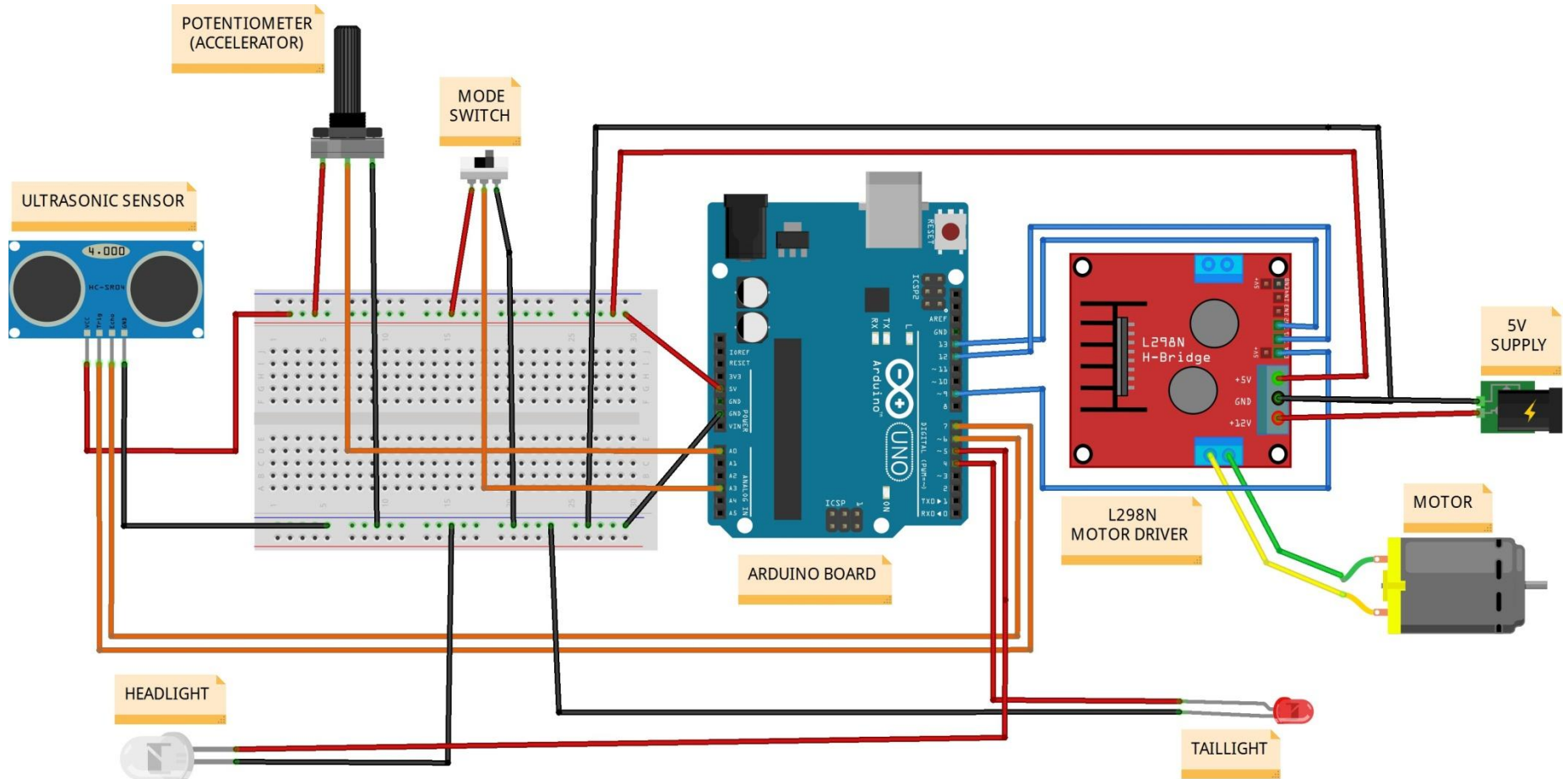
- Controls power delivered to electric motor to ensure optimal vehicle performance.
- Can be re-programmed by uploading updated code or replacing chips.

ARDUINO UNO

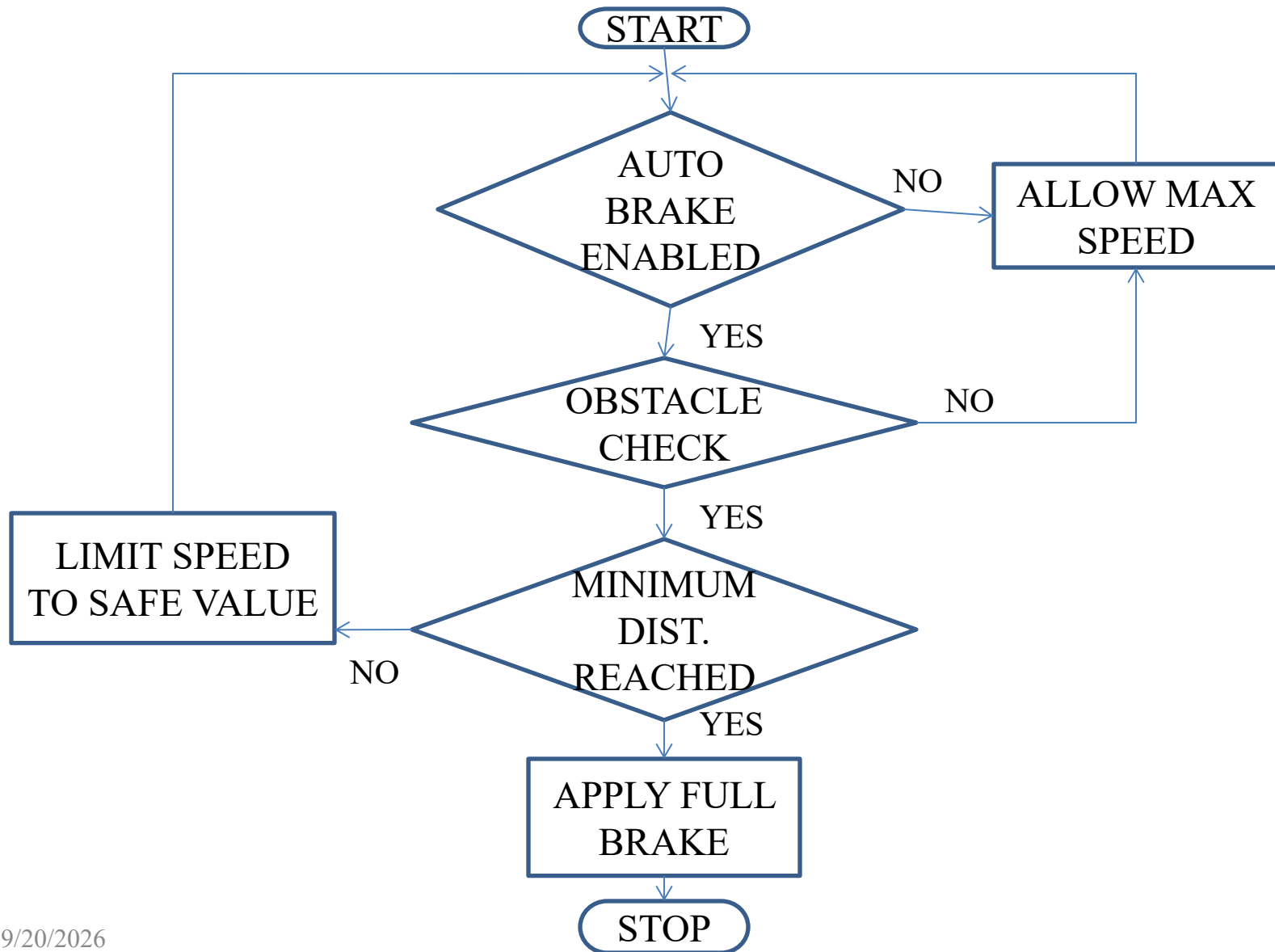
- Microcontroller board based on the ATmega328
- 14 digital input/output pins (of which 6 can be used as PWM outputs)
- 6 analog inputs
- Uses 16 MHz crystal oscillator
- USB support

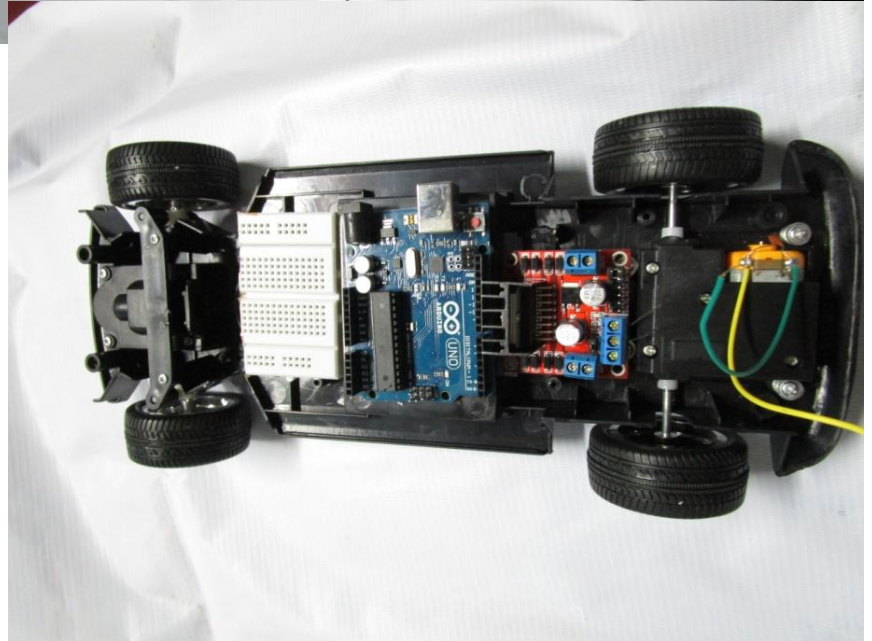
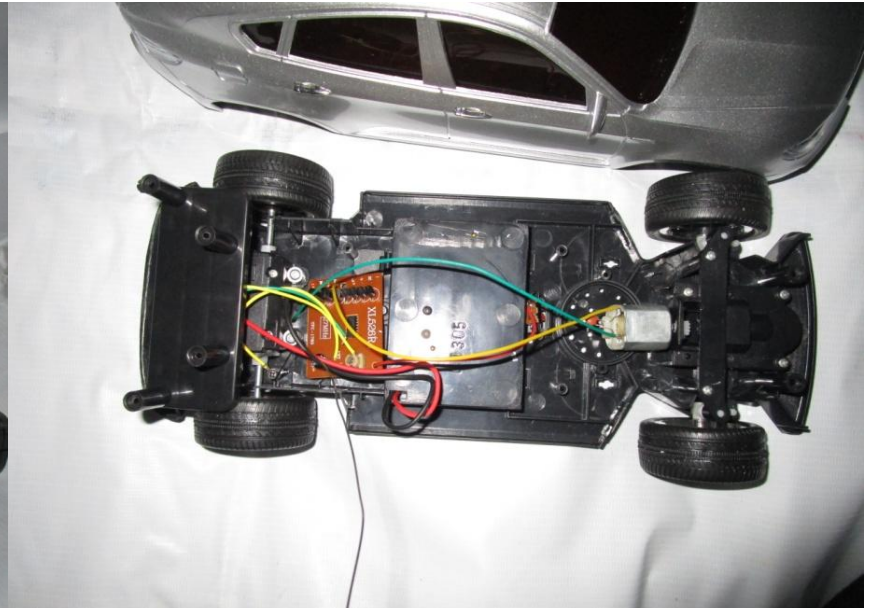


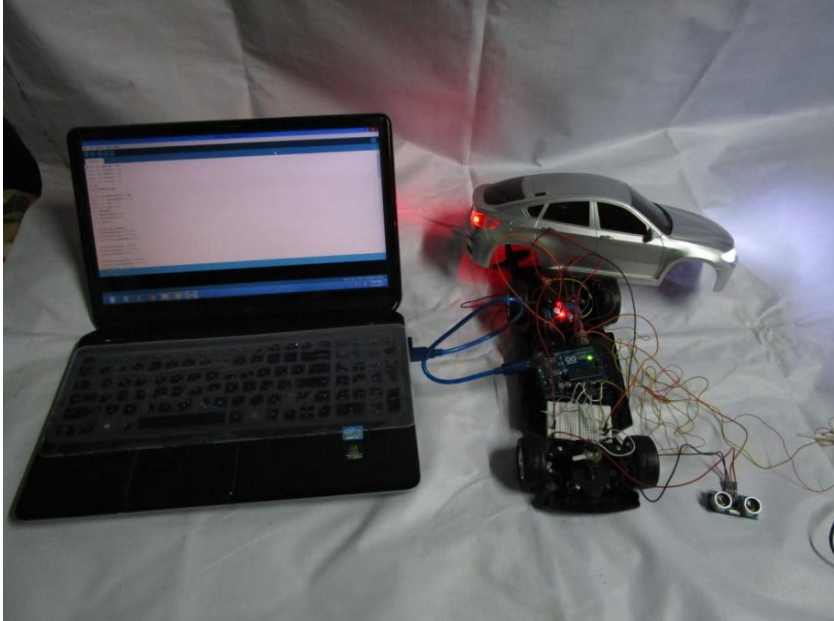
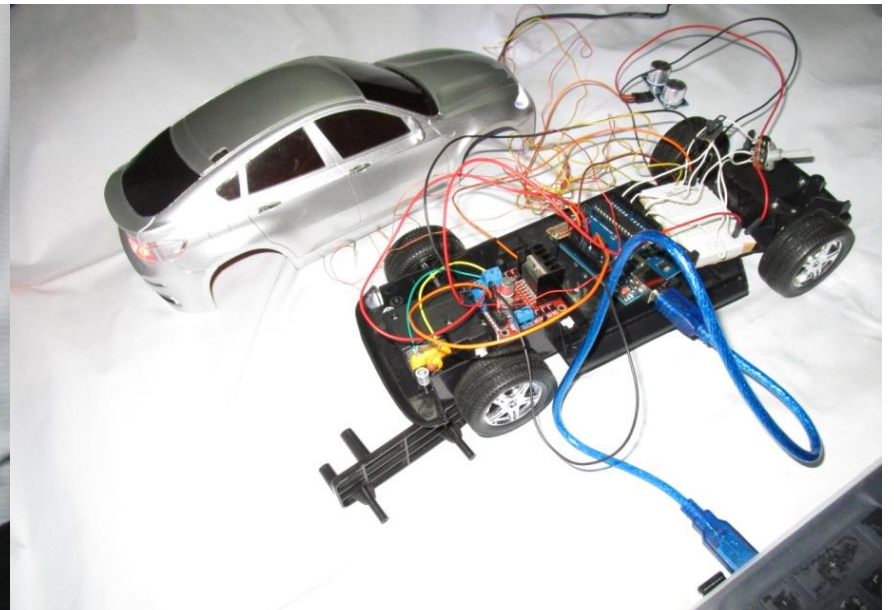
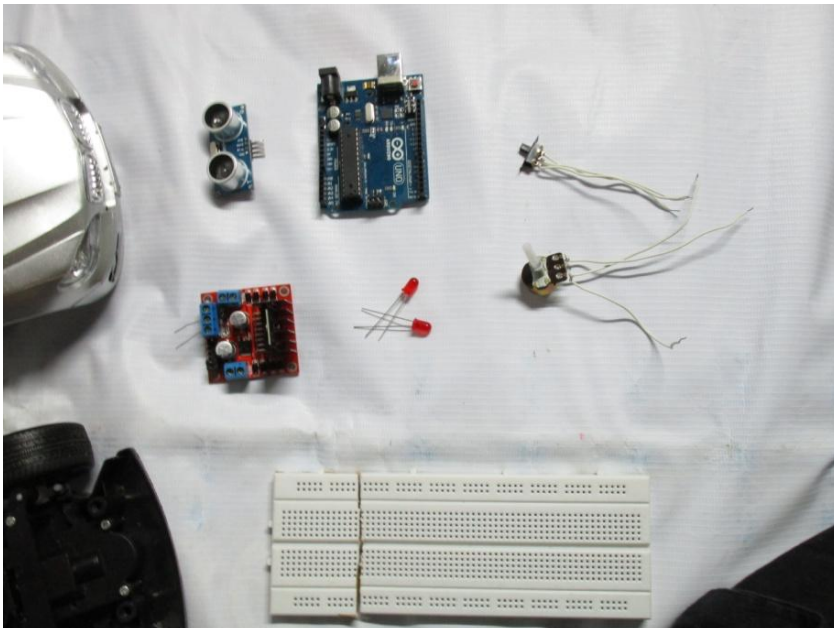
CIRCUIT DIAGRAM



PROGRAM FLOWCHART

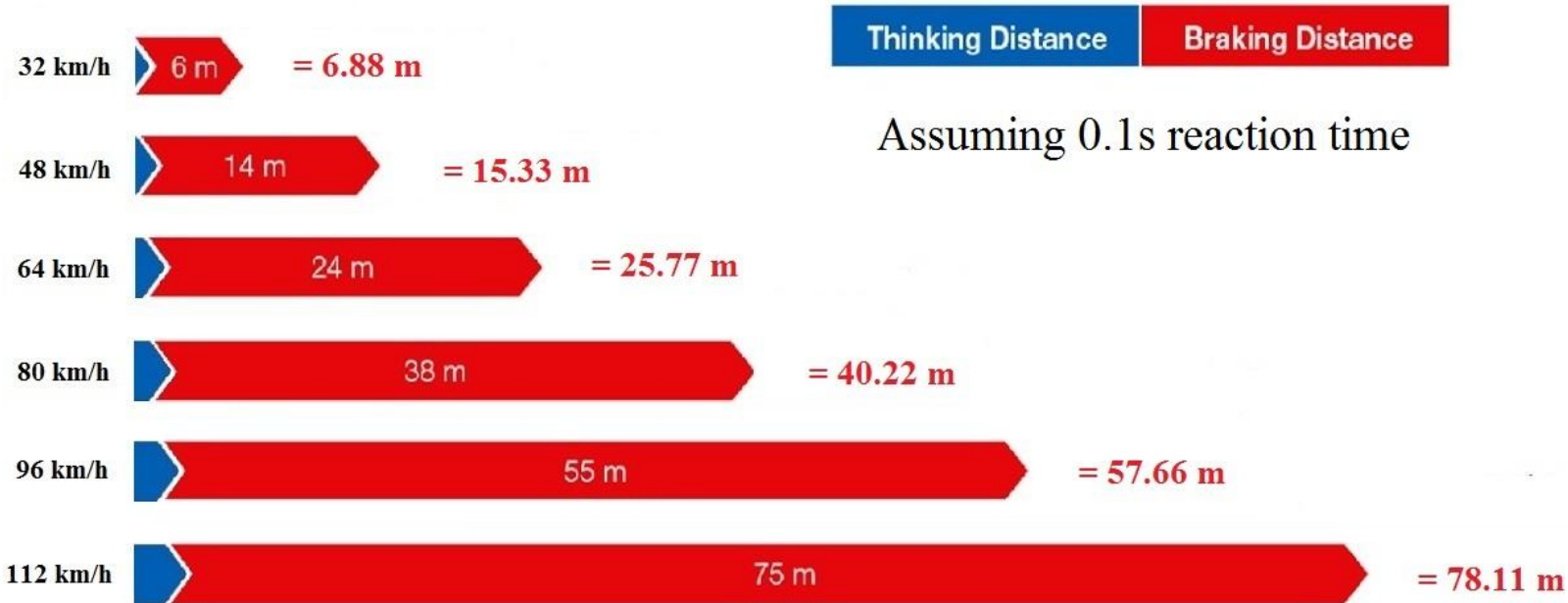






IMPROVED STOPPING DISTANCE

IMPROVED STOPPING DISTANCE



PERCENTAGE DECREASE IN STOPPING DISTANCE

SPEED	HUMAN THINKING DIST.	SYSTEM THINKING DIST.	BRAKING DIST.	% DECREASE IN TOTAL BRAKING DIST.
km/h	m	m	m	%
32	6	0.89	6	42.59
48	9	1.33	14	33.33
64	12	1.78	24	28.40
80	15	2.22	38	24.11
96	18	2.67	55	21.00
112	21	3.11	75	18.63

HARDWARE PROTOTYPE COST

ITEM	PRICE
TOY CAR	1190
ARUDUINO UNO BOARD	550
L298N DRIVER	300
HC-SR04 SENSOR	200
CONNECTORS	65
ENCLOSURE BOX	45
OTHER COMPONENTS	20
TOTAL	2370

SOFTWARES USED



ADVANTAGES

- Stopping distance of the vehicle can be considerably reduced as the system has very fast response time.
- It can avoid rear end collisions of vehicles due to driver negligence.
- In the case of unavoidable collisions the system reduces the degree of damage.

ADVANTAGES

- Based on the traffic rules of each country, the system may be modified.
- In control engineering aspect,the behaviour of the system can be change by modifying the program uploaded to the electronic control unit.

DISADVANTAGES

- The conditions and infrastructure of the roads in each country will differ from one another, depending on these conditions our system will have to be modified.
- Its not necessary that all obstacles standing in front of our vehicle need to be considered as a threat, and even in those cases our system will apply brakes to the vehicle.
- The limitations of the sensor techonology used will become a disadvantage for the system.

FUTURE SCOPE

- Rain sensors and day night sensors can sense the environment conditions the visibility and road conditions. In wet road conditions the friction coefficient between the tyres and the road decreases so the system can automatically further reduce the speed to new safe values.
- Seat belt pretensioners can be connected to the system. Pretensioner is to tighten up any slack in the seatbelt in the event of a crash to make sure a seatbelt restrains an occupant as early as possible in a crash.

CONCLUSIONS

- The advent of technology mainly focuses on reducing human effort in every field. This system may be considered as a step to reduce the mental and physical efforts taken by the driver to safely drive the vehicle.
- Even if our system may not prevent collisions in every critical scenario, it will definitely reduce the impact of collisions preventing loss of life.

- By the proper implementation of this system, the number of accidents occurring every year can be reduced by a considerable amount.
- As the system can be made flexible, it can be suited for all kinds of vehicles and to the different traffic rules existing in different countries.


```
const int modesw = A3;
const int trigPin = 7;
const int echoPin = 6;
const int headlgt = 4;
const int taillgt = 5;
int m;
long duration,cm;
const int analogInPin = A0;
const int analogOutPin = 9;
const int in1 = 12;
const int in2 = 13;
int safespd;
int sensorValue = 0;
int outputValue = 0;
void setup()
{
  pinMode(headlgt, OUTPUT);
  pinMode(taillgt, OUTPUT);
  digitalWrite(headlgt,HIGH);
  pinMode(echoPin, INPUT);
  pinMode(trigPin, OUTPUT);
  pinMode(modesw,INPUT);
  pinMode(analogInPin, INPUT);
  pinMode(analogOutPin, OUTPUT);
  pinMode(in1, OUTPUT);
  pinMode(in2, OUTPUT);
}
```

```
void loop()
{
  m=digitalRead(modesw);
  if(m==0)
  {
    digitalWrite(in1, HIGH);
    digitalWrite(in2, LOW);
    sensorValue = analogRead(analogInPin);
    outputValue = map(sensorValue, 0, 1023, 0, 255);
    analogWrite(analogOutPin, outputValue);
  }
  else if(m==1)
  {
    digitalWrite(in1, HIGH);
    digitalWrite(in2, LOW);

    digitalWrite(trigPin, LOW);
    delayMicroseconds(2);
    digitalWrite(trigPin, HIGH);
    delayMicroseconds(10);
    digitalWrite(trigPin, LOW);

    duration = pulseIn(echoPin, HIGH);
    cm = microsecondsToCentimeters(duration);
```

```
if(cm<20)
{
    rept:
    digitalWrite(in1, LOW);
    digitalWrite(in2, LOW);
    analogWrite(analogOutPin, 255);
    digitalWrite(taillgt,HIGH);
    goto rept;
}
else if(cm<100)
{
    safespd=50;

}
else if(cm<170)
{
    safespd=100;
}
else if(cm<200)
{
    safespd=150;
}
else if(cm<250)
{
    safespd=180;
}
```

```
else if(cm<300)
{
safespd=225;
}
else
{
safespd=255;
}
sensorValue = analogRead(analogInPin);
outputValue = map(sensorValue, 0, 1023, 0, safespd);
analogWrite(analogOutPin, outputValue);
}
delay(50);
}
long microsecondsToCentimeters(long microseconds)
{
return microseconds / 29 / 2;
}
```

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THANK YOU